DRUGS · CORTICOSTEROIDS

Methylprednisolone and hydrocortisone

Hydrocortisone and methylprednisolone are the intravenous corticosteroids of hospital practice. Hydrocortisone is used for adrenal crisis and perioperative steroid cover in patients on long-term steroids, and as an adjunct in severe allergic reactions. Methylprednisolone is used as pulsed high-dose therapy for vasculitis such as granulomatosis with polyangiitis, autoimmune inner ear disease and other severe inflammatory diseases, and through the drum.

1 CHAPTER ONE At a glance

The drugs on one page.

1.1 Summary

HYDROCORTISONE AND METHYLPREDNISOLONE

	SUMMARY
Agents	Hydrocortisone (short-acting, with mineralocorticoid effect); methylprednisolone (intermediate-acting, potent, little mineralocorticoid effect)
ENT uses	Adrenal crisis; perioperative steroid cover; adjunct in anaphylaxis and angioedema; methylprednisolone pulses for severe GPA, autoimmune inner ear disease and severe inflammatory airway disease; intratympanic methylprednisolone for Ménière's disease
Adrenal crisis	Hydrocortisone 100 mg intravenously or intramuscularly at once, then 200 mg over 24 hours by infusion (or 50 mg every 6 hours), with fluids
Perioperative cover	For patients on 5 mg or more of prednisolone (or equivalent) for 4 weeks or more: hydrocortisone 100 mg at induction, then for major surgery 200 mg over 24 hours until eating, then double the usual oral dose for 48 hours (minor surgery: usual dose plus 100 mg at induction, then double dose for 24 to 48 hours)
Pulses	Methylprednisolone 500 mg to 1 g intravenously daily for 3 days

	SUMMARY
Child dose	Hydrocortisone in adrenal crisis 2 mg/kg (maximum 100 mg) then as advised; methylprednisolone pulses 10 to 30 mg/kg (maximum 1 g)

CHAPTER TWO

2 Using them in ENT

Indications.

2.1 The patient on long-term steroids

Any patient on long-term steroids, or who has stopped them within the past 3 months, may not mount a stress response. Surgery, vomiting or severe infection can precipitate an **adrenal crisis**: hypotension, hyponatraemia, hypoglycaemia and collapse. Plan perioperative cover with the anaesthetist (national guidance from the Association of Anaesthetists and the Society for Endocrinology) and give hydrocortisone at once to any such patient who is unwell.

2.2 Vasculitis and autoimmune ear disease

Severe or organ-threatening **granulomatosis with polyangiitis** (subglottic stenosis with airway compromise, orbital disease, rapidly progressive renal disease) is induced with methylprednisolone pulses and rituximab or cyclophosphamide under rheumatology. **Autoimmune inner ear disease** (rapidly progressive bilateral sensorineural loss) is treated with high-dose prednisolone (1 mg/kg for 4 weeks), sometimes after an intravenous pulse.

2.3 Anaphylaxis

UK resuscitation guidance no longer recommends routine steroids in the initial treatment of anaphylaxis: adrenaline and fluids come first. Steroids may be given for refractory reactions or asthma.

3 CHAPTER THREE Safety

What goes wrong.

3.1 Adverse effects

As for all glucocorticoids, magnified by pulses: hyperglycaemia, hypertension, fluid retention (hydrocortisone), psychosis, infection, arrhythmia with rapid pulses (infuse methylprednisolone over at least 30 minutes), avascular necrosis.

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- Who needs **perioperative steroid cover** and the regimen.
- Recognition and treatment of **adrenal crisis**.
- Pulsed methylprednisolone in GPA and its indications.

4.2 Common pitfalls

COMMON PITFALLS

- Missing a patient's steroid history before a major head and neck operation.
- Delaying hydrocortisone in a collapsed patient on long-term steroids.

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